Lecture 02
Formal Foundations of Computing

Algorithms, AI & Data Science II

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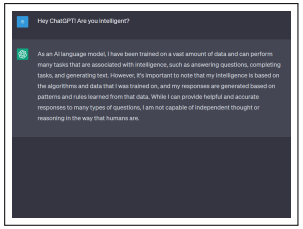
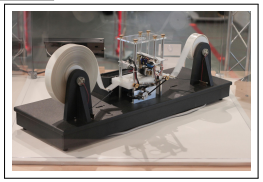
Motivation

- deep understanding of algorithms and artificial intelligence requires to **formalize "computation"**
- how can we **formally state a problem** such that a machine can given an answer?
- we explore **formal languages** and study **automata models** to address **word decision problems**
- we define **Turing machines** and investigate **computability and "machine intelligence"**
- we start with basic **mathematical foundations of computer science**
  1. set theory and set operations
  2. relations and functions
  3. notational conventions



response from large language model ChatGPT

image credit: OpenAI screenshot, taken on 12/04/2023



Physical model of a **Turing machine**

image credit: Rocky Acosta, Wikimedia Commons, CC-BY-SA

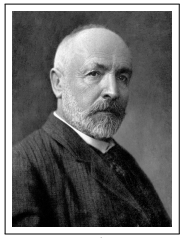
Notes:

- **L02: Formal Foundations of Computing** 30.04.2025
- **Educational objective:** We introduce foundational mathematical concepts like sets and relations, study formal languages and grammars and introduce the Chomsky hierarchy of formal languages.
  - Basic Set Theory
  - Relations and Functions
  - Formal Languages and Grammars
  - Chomsky Hierarchy of Formal Languages
- **Handout version:** *April 30, 2025*

Notes:

Set theory

- **set theory** = mathematical study of relationships between **collections of objects**
- important foundation of discrete mathematics, formal logics and computer science → G Cantor, 1874
- sets are **unordered** collections of **distinguishable** objects
- sets can be **finite or infinite**
- we can unambiguously determine whether an **object is element of a set or not**



Georg Cantor (1845 – 1918)

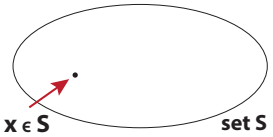


image credit: Wikimedia Commons, public domain

Foundational set theory

definitions

- A **set** is an unordered collection of distinguishable objects, which are called **elements** or **members** of the set.
- For any object x we can unambiguously determine whether it is element of a set S. If x is member of S we write $x \in S$ and $x \notin S$ otherwise.
- The **cardinality** $|S|$ is defined as the number of elements in S.
- We denote the **empty set** (without elements) as $\emptyset$ with $|\emptyset| = 0$.
- We can define sets in (finite or infinite) **enumeration notation**, i.e. $S = \{e_1, e_2, e_3, \ldots\}$.
- We can define sets in **set-builder notation**, i.e. $\{x|p(x)\}$ where p is a condition (or logical predicate) that holds for all members $x \in S$
- S_1 **is subset of** S_2 ($S_1 \subseteq S_2$) iff for each $x \in S_1$ we have $x \in S_2$
- S_1 **and** S_2 **are equal** ($S_1 = S_2$) iff for each $x \in S_1$ we have $x \in S_2$ and vice-versa, i.e. $S_1 \subseteq S_2$ and $S_2 \subseteq S_1$
- S_1 **is proper subset of** S_2 ($S_1 \subset S_2$) iff for each $x \in S_1$ we have $x \in S_2$ and $S_1 \neq S_2$

examples

- $S = \{1, \text{'Cat'}, 42.5\}$ is a set with $|S| = 3$
- $S = \{1, 2, 3, 2\}$ is not a valid set
- $\mathbb{N} = \{1, 2, 3, 4, \ldots\}$ is an (infinite) set
- $\mathbb{N} = \{x | x \text{ is natural number }\}$
- $S_1 = \{1, 2, 3\} \subseteq \{1, 2, 3, 4\} = S_2$
- $\emptyset \subseteq S$ for any set S
- $S_1 = \{1, 2, 3\} = \{3, 2, 1\} = S_2$
- $S_1 = \{1, 2, 3\} \subset \{1, 2, 3, 4\} = S_2$

Notes:

Notes:

Set operations, complement, and power set

definitions

Let S_1 and S_2 be arbitrary sets. We define the following binary operations:

- $S_1 \cup S_2 := \{x | x \in S_1 \text{ or } x \in S_2\}$ **union**
- $S_1 \cap S_2 := \{x | x \in S_1 \text{ and } x \in S_2\}$ **intersection**
- $S_1 - S_2 := \{x | x \in S_1 \text{ and } x \notin S_2\}$ **difference**
- $S_1 \triangle S_2 := (S_1 \cup S_2) - (S_2 \cap S_2) = \{x | x \in S_1 \text{ xor } x \in S_2\}$ **symmetric difference**
- $S_1 \times S_2 := \{(x, y) | x \in S_1 \text{ and } y \in S_2\}$ **product**

- for given set S we define **complement** S^C (relative to some "universe") as the set of all elements in the universe that are not element of S
- for given set S we define **power set** 2^S as the set of all subsets of S, i.e. $2^S := \{S' : S' \subseteq S\}$

examples

Let $S_1 = \{1, 2, 3\}$ and $S_2 = \{2, 3, 4, 5\}$.

- $S_1 \cup S_2 = \{1, 2, 3, 4, 5\}$
- $S_1 \cap S_2 = \{2, 3\}$
- $S_1 - S_2 = \{1\}$
- $S_1 \triangle S_2 = \{1, 4, 5\}$
- $S_1 \times S_2 = \{(1,2), (1,3), (1,4), (1,5), (2,2), (2,3), (2,4), (2,5), (3,2), (3,3), (3,4), (3,5)\}$

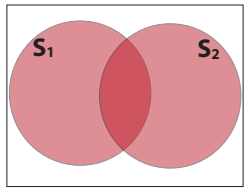
  with $|S_1 \times S_2| = 12$
- For $S = \{1, 2, 3\}$ we have

  $2^S = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1,2\}, \{1,3\}, \{2,3\}, \{1,2,3\}\}$

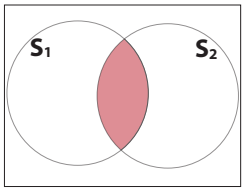
  with $|2^S| = 2^{|S|} = 8$

Venn diagrams

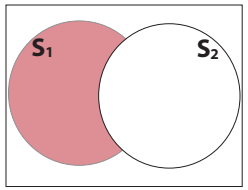
we can use **Venn diagrams** to visually represent operations on sets



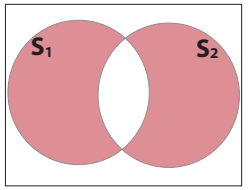
$S_1 \cup S_2$



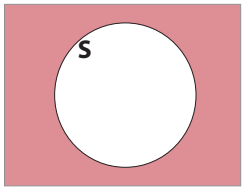
$S_1 \cap S_2$



$S_1 - S_2$



$S_1 \triangle S_2$



S^C

Notes:

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Relations

- elements of sets often exhibit **relationships** that we can capture mathematically

definition

- Let S_1 and S_2 be arbitrary sets. A (binary) relation R between S_1 and S_2 is a subset of the product set $S_1 \times S_2$, i.e. $R \subseteq S_1 \times S_2$ and $R \in 2^{S_1 \times S_2}$
- Instead of $(a, b) \in R$ we can write $R(a, b)$ or aRb.

- relations $R \subseteq S_1 \times S_2$ can be between elements of **different sets** $S_1 \neq S_2$
- we can define relations $R \subseteq S \times S$ between elements of a **single set**, in which case we call R a **relation on** S

examples

- For $S_1 = \{\text{Peter}, \text{Anna}\}$ and $S_2 = \{\text{C++}, \text{python}\}$

$$R = \{(\text{Peter}, C++), (\text{Anna}, \text{python}), (\text{Anna}, \text{C++})\}$$

  is a relation
- $\emptyset$ is an (empty) relation between any pair of sets
- comparator $<$ is a relation on natural numbers $\mathbb{N}$ with $(2, 5) \in <$ and $(5, 2) \notin <$
- $R = \{(x, y) \in \mathbb{N} \times \mathbb{N} | y = x^2\}$ is a relation on natural numbers $\mathbb{N}$

Properties of relations

definitions

- relation R on S is **reflexive** iff xRx for any $x \in S$
- relation R on S is **symmetric** iff for all $x, y \in S$ with xRy we also have yRx
- relation R on S is **transitive** iff for all $x, y, z \in S$ with xRy and yRz we also have xRz
- a relation R between sets X, Y is called **functional** (or right-unique) iff for all $x \in X$ and $y, y' \in Y$ with $R(x, y)$ and $R(x, y')$ we have $y = y'$
- relation R between X, Y is called **injective** (or left-unique) iff for all $x, x' \in X$ and $y \in Y$ with $R(x, y)$ and $R(x', y)$ we have $x = x'$

- we call functional relation $R \subseteq X \times Y$ a **function** $R : X \to Y$ and write $R(x) = y$ instead of $R(x, y)$
- we call a functional relation $R \subseteq X \times Y$ that is injective **one-to-one mapping**

examples

- comparator $<$ on $\mathbb{N}$ is neither reflexive nor symmetric
- comparator $\leq$ on $\mathbb{N}$ is reflexive but not symmetric
- $<$ and $\leq$ on $\mathbb{N}$ are transitive
- $R = \{(x, x), (y, y), (z, z), (y, z), (z, y)\}$ is reflexive, symmetric, and transitive
- $<$ and $\leq$ on $\mathbb{N}$ are not functional
- $R(x) = x^2$ is functional
- $R(x) = x^2$ is not injective
- $R(x) = x + 1$ is a one-to-one mapping that maps every $x \in \mathbb{N}$ to $x + 1$

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Operations on relations

- Let $R \subseteq S_1 \times S_2$ be a relation. The **inverse of** R is defined as

$$R^{-1} := \{(y, x) \in S_2 \times S_1 | (x, y) \in R\}$$

- Let $R_1 \subseteq S_1 \times S_2$ and $R_2 \subseteq S_3 \times S_4$ be relations. The **inner product of** R_1 **and** R_2 is defined as

$$R_1 \otimes R_2 := \{((a, c), (b, d)) \in (S_1 \times S_3) \times (S_2 \times S_4) | aR_1b \text{ and } cR_2d\}$$

- Let $R_1 \subseteq S_1 \times S_2$ and $R_2 \subseteq S_2 \times S_3$ relations. The **composition of** R_1 **and** R_2 is defined as

$$R_1 \circ R_2 := \{(a, c) \in (S_1 \times S_3) | aR_1b \text{ and } bR_2c \text{ for some } b\}$$

Group Exercise 02-01

1. Consider the "smaller than" relation $<$ on $\mathbb{R}$ and let $<^{-1}$ be the inverse of $<$. Provide a set builder definition for the elements in the inner product $< \otimes <^{-1}$. Give an example for elements of the resulting relation.

2. Prove that $(R \cup S)^{-1} = R^{-1} \cup S^{-1}$

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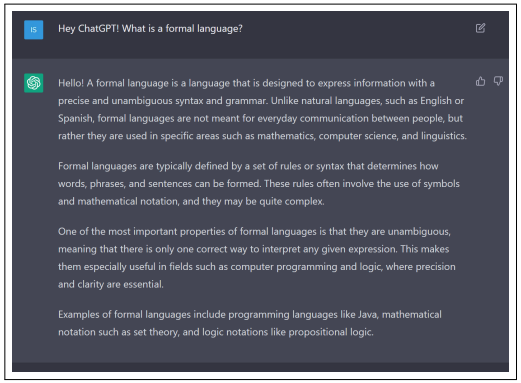
Languages and computers

- humans use **natural languages** (like German, English, French) to communicate
- in natural language, meaning is often **ambiguous or depends on context**, e.g.

  "She saw the man with the telescope."
- to communicate with machines, we (generally) **formalize languages** such that the meaning of statements is precisely defined

examples of formal languages

1. programming languages (C, C++, python)
2. arithmetic expressions
3. Boolean algebra → L03 - Propositional Logic and Boolean Algebra



output of large language model **ChatGPT**

image credit: OpenAI screenshot, created 29/03/2023

What is a formal language?

definition

Let Σ be a **finite alphabet**, i.e. a set of symbols.

- We call a finite sequence $w \in \Sigma^i$ of i symbols a **word** with length $|w| := i$. We denote the set of all words of length i as Σ^i.
- We call ϵ the **empty word** (with length zero).
- We use w^R to denote the word that we get by reversing the sequence of symbols.
- With $\Sigma^0 = \{\epsilon\}$ we recursively define

$$\Sigma^{i+1} := \{w | w = xy \text{ with } x \in \Sigma^i \text{ and } y \in \Sigma\}$$

- We define $\Sigma^* := \bigcup_{i \geq 0} \Sigma^i$ and $\Sigma^+ := \bigcup_{i > 0} \Sigma^i$
- A **formal language** $L \subseteq \Sigma^*$ is a (possibly infinite) set of words over alphabet Σ.

examples

- $L = \{aa, bb, abba\}$ is a (finite) formal language over alphabet $\Sigma = \{a, b\}$
- $L = \{x \in \Sigma^* | x = x^R\}$ is an (infinite) formal language of palindroms over alphabet $\Sigma = \{a, b\}$
- $L = \{x | x \text{ is syntactically correct python program}\}$ is a formal language over the alphabet of ASCII characters

how can we define **syntax** ("$\sigma\upsilon\nu - \tau\alpha\xi$") of words of a formal language L?

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Grammars

- we can use **grammars** to define formal languages based on **production rules**

definition

A **grammar** G is a four-tuple $G = (V, \Sigma, P, S)$ where

- V is a finite set of **variables** used in production rules,
- Σ is a finite alphabet of **terminal symbols** with $V \cap \Sigma = \emptyset$,
- P is a finite set of **production rules** with $P \subseteq (V \cup \Sigma)^+ \times (V \cup \Sigma)^*$, and
- $S \in V$ is a **start symbol**.

For $(X, Y) \in P$ we write $X \to Y$ to denote that X can be substituted by Y.
We use $X \to^* Y$ to denote that there is a (possibly transitive) application of production rules that allows to substitute X by Y.
We say that $L(G) = \{w \in \Sigma^* | S \to^* w\}$ is the **language generated by grammar** G.

- P is a relation that is **not necessarily functional**, i.e. we can have $X \to Y \in P$ and $X \to Y' \in P$, which we denote as $X \to Y \mid Y'$

example

$G = (\{A, B\}, \{a, b\}, P, S)$ with

$$P = \{S \to AB,\\ A \to aA \mid \epsilon,\\ B \to bB \mid \epsilon \}$$

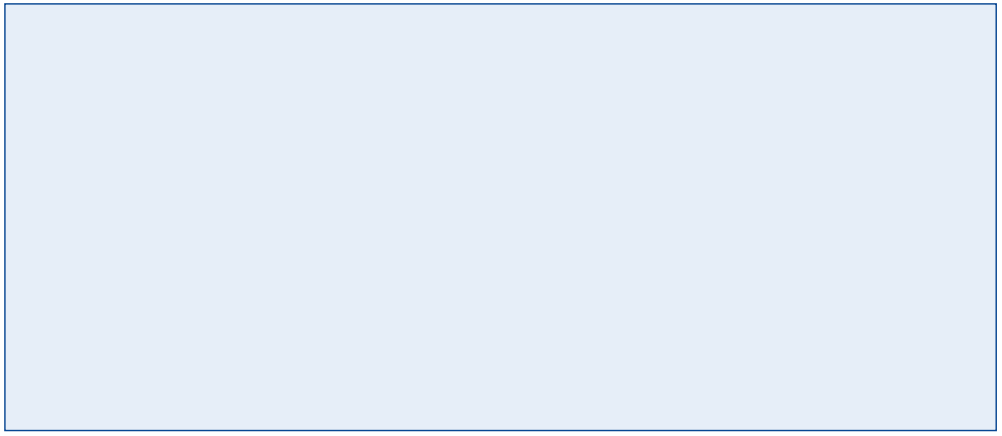
Starting with S we can apply the following substitutions:

$$S \to AB\\ AB \to aAB\\ aAb \to aaAB\\ aaAB \to aaAbB\\ aaAbB \to aabB\\ aabB \to aab$$

Due to $S \to^* aab$ we have $aab \in L(G)$.

Group Exercise 02-02

Write a grammar for the formal language $L = \{a^n b^n c^n | n \geq 1\}$. Explain how the production rules are used to produce the word *aaabbbccc*.



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Chomsky hierarchy

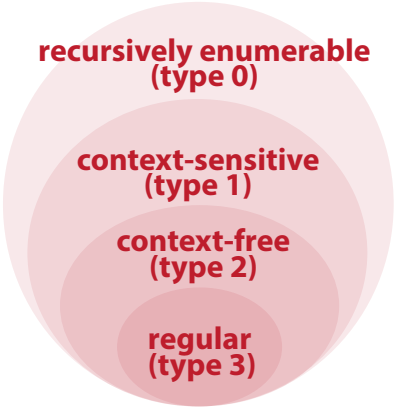
- we **categorize grammars into a hierarchy** based on constraints that must be satisfied by production rules

definition

Let $G = (V, \Sigma, P, S)$ be a grammar.

- Every grammar G is **recursively enumerable (type 0)**
- Grammar G is **context-sensitive (type 1)**, iff for all production rules $w_1 \to w_2 \in P : |w_1| \leq |w_2|$ (or rule has the form $S \to \epsilon$)
- Grammar G of type 1 is **context-free (type 2)**, iff for all rules $w_1 \to w_2 \in P : w_1 \in V$
- Grammar G of type 2 is **regular (type 3)**, iff for all rules $w_1 \to w_2 \in P : w_2 \in \Sigma \cup \Sigma V \cup \{\epsilon\}$

We say that a formal language $L \subseteq \Sigma^*$ is of type 0, 1, 2 or 3 iff a type 0, 1, 2 or 3 grammar G exists with $L(G)$, respectively.



Venn diagram of Chomsky hierarchy

- higher type = stricter rules = **less expressive**

Group Exercise 02-03

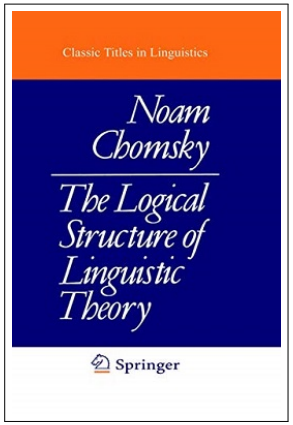
Consider the grammar from group exercise 02-02 and determine its type according to the Chomsky hierarchy.

Notes:

Notes:

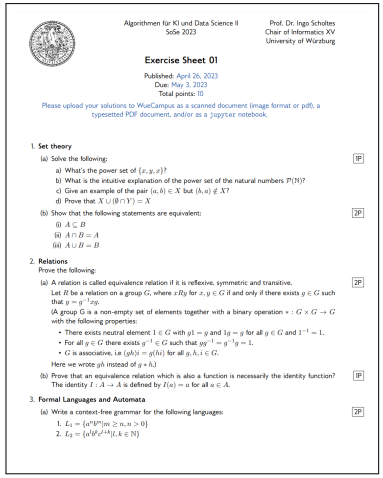
In summary …

- we have covered **formal foundations of computing**
- we introduced **formal languages**, which are defined as (possibly infinite) sets of words over finite alphabet
- we can specify the syntax of formal languages based on **grammars with production rules**
- we use the **Chomsky hierarchy** to categorize grammars (and the languages that they produce) based on their production rules
- formal languages allow us to formalize **which problems are decidable** by an algorithm
- how can we **algorithmically decide** whether a given word w is in a language L?



Exercise sheet 01

- **first exercise sheet** will be released next Monday
- solutions are due **May 12th** (via WueCampus)
- present your solution to earn bonus points



Algorithmen für KI und Data Science II
SoSe 2023

Prof. Dr. Ingo Scholtes
Chair of Informatics XV
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Exercise Sheet 01

Published: April 26, 2023
Due: May 3, 2023
Total points: 10

Please upload your solutions to WueCampus as a scanned document (image format or pdf), a typesetted PDF document, and/or as a Jupyter notebook.

1. **Set theory**
   (a) Solve the following: [1P]
   a) What's the power set of $\{x, y, x\}$?
   b) What is the intuitive explanation of the power set of the natural numbers $\mathcal{P}(\mathbb{N})$?
   c) Give an example of the pair $(a, b) \in X$ but $(b, a) \notin X$?
   d) Prove that $X \cup (\emptyset \cap Y) = X$
   (b) Show that the following statements are equivalent: [2P]
   (i) $A \subseteq B$
   (ii) $A \cap B = A$
   (iii) $A \cup B = B$
2. **Relations**
   Prove the following:
   (a) A relation is called equivalence relation if it is reflexive, symmetric and transitive. [2P]
   Let R be a relation on a group G, where $x R y$ for $x, y \in G$ if and only if there exists $g \in G$ such that $y = g^{-1} x g$.
   (A group G is a non-empty set of elements together with a binary operation $\kappa : G \times G \to G$ with the following properties:
   - There exists neutral element $1 \in G$ with $g1 = g$ and $1g = g$ for all $g \in G$ and $1^{-1} = 1$.
   - For all $g \in G$ there exists $g^{-1} \in G$ such that $gg^{-1} = g^{-1}g = 1$.
   - G is associative, i.e $(gh)i = g(hi)$ for all $g, h, i \in G$.

   Here we wrote gh instead of $g * h$.)
   (b) Prove that an equivalence relation which is also a function is necessarily the identity function? [1P]
   The identity $I : A \to A$ is defined by $I(a) = a$ for all $a \in A$.
3. **Formal Languages and Automata**
   (a) Write a context-free grammar for the following languages: [2P]
   1. $L_1 = \{a^n b^m | m \geq n, n > 0\}$
   2. $L_2 = \{a^l b^k c^{l+k} | l, k \in \mathbb{N}\}$

Notes:

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Self-study questions

1. Explain the difference between a relation and a function.
2. Give an example for a relation R on $\mathbb{N}$ that is reflexive, symmetric, and transitive.
3. Show that your relation R from above defines a partition of $\mathbb{N}$, i.e. a partition of $\mathbb{N}$ in pair-wise non-overlapping subsets of $\mathbb{N}$.
4. Give an example for a relation R that is not functional but injective.
5. Provide the formal definition of a grammar. What is the difference between V and Σ in the definition?
6. Give an example for a grammar where the production rules never terminate (thus not a language without any words).
7. Give a grammar for the language of all bit strings that correspond to even natural numbers.
8. Using the production rules of the previous question, test whether a large language model like ChatGPT can correctly derive the resulting language.
9. What is the difference between a content-sensitive and context-free language?
10. Give an example for a language that is regular.
11. Write a grammar for the language of correctly bracketed arithmetic formulas with arithmetic operations $+$ and $*$. For simplicity consider formulas with two variables a, b only.

References

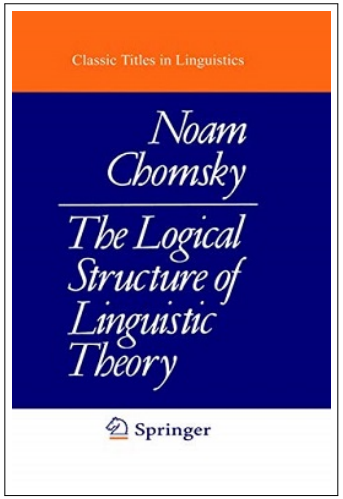
reading list

- G Cantor: **Ueber eine Eigenschaft des Inbegriffes aller reellen algebraischen Zahlen**, Journal über die Reine und Angewandte Mathematik, 1874
- N Chomsky: **Three models for the description of language**, 1956
- N Chomsky: **The Logical Structure of Linguistic Theory**, 1975

acknowledgments

some examples are based on the following books and lectures:

- U Schöning: **Theoretische Informatik - kurzgefasst**, Spektrum Akadem. Verlag, 2001
- J Esparza: **Automata theory: An algorithmic approach**, lecture notes, 2020



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